Vijay Babubhai Doba

Erlangen, Germany | vijaydoba0011@gmail.com | +49 178 579 3147
LinkedIn: vijaydoba | GitHub: vijaydoba | Portfolio: vijaydoba.github.io/Portfolio



PROFILE

M.Sc. Artificial Intelligence student at FAU Erlangen–Nürnberg who builds **LLM and RAG applications end to end**. Shipped **FilingIQ** (Python, FastAPI, Claude API, ChromaDB): a retrieval assistant that answers over financial filings only from cited sources and holds back when it cannot find one. Research background in **out-of-distribution detection and applied ML (PyTorch)**, including a CVPR 2024 reproducibility study where a CLIP-based OOD detector reached **AUROC 98.63** on Places. Comfortable owning a feature from data pipeline and model through API and UI. Seeking a **junior AI / LLM Engineer** role. English C1, German A2 (improving).

EDUCATION

M.Sc. Artificial Intelligence — FAU Erlangen–Nürnberg Apr 2023 – Present (expected 2027)
Erlangen–Nürnberg, Germany

- Relevant coursework: Machine Learning, Artificial Intelligence, Data Science, Business Intelligence, Advanced Programming Techniques, HCI.

B.E. Computer Engineering — Government Engineering College, Bhavnagar Jul 2018 – Apr 2022
Bhavnagar, India — GPA 7.76 / 10.0

- Coursework: Data Structures, Algorithms, Database Systems, OOP, Computer Networks, IT Security, AI/ML, Software Engineering.

PROJECTS

FilingIQ — Financial Document RAG Assistant 2026
Python, FastAPI, Anthropic Claude API, ChromaDB, RAG, Docker

- Built a multi-company **RAG** pipeline: auto-fetches SEC **EDGAR** filings, chunks/embeds locally, and persists vectors in **ChromaDB**.
- Implemented citation-grounded Q&A via the **Claude API**, a two-company comparison mode, and streaming responses.
- Extended coverage to India (NSE) and Germany (XETRA) filings via fuzzy company search; containerized with **Docker**.

Reproducibility Study — 6 CVPR 2024 Papers 2025
Python, PyTorch, CLIP, CUDA, Linux

- Reproduced **6 CVPR 2024 papers** end-to-end (OOD detection, federated learning, adversarial robustness, binary neural nets, stereo matching) from undocumented repos — fixed missing scripts/hyperparameters, dataset access issues, and setup docs across all six.
- Debugged a missing Dirichlet-partitioning parameter in **FedAS** and tuned **LANDER**'s training schedule, beating both papers' reported accuracy (**91.03% vs. 89.79%**; **53.34% vs. 52.60%**); adapted a CLIP-based OOD model (ID-like) via mixed-precision training to **AUROC 98.63** on Places.

Chatrio — Real-time Anonymous Chat Platform 2025
React, TypeScript, Node.js, Express, Socket.IO, PWA

- Built full-stack real-time messaging over WebSockets (Socket.IO + Node.js) with interest matching, typing indicators, and image sharing; deployed as a cross-device PWA.

PROFESSIONAL EXPERIENCE

Web Developer — iFLUTTER May 2022 – May 2023
Surat, India (Hybrid)

- Built and maintained custom client websites end-to-end — development, testing, and QA — delivering to specification.
- Structured modular, maintainable codebases and kept technical documentation current across multiple projects.
- Diagnosed and resolved production issues; optimized site performance and load behaviour.

Backend Intern (Laravel) — Spider Technology Jan 2022 – Apr 2022
Surat, India (Hybrid)

- Built a **Sales Lead Automation & Task Control System** (B.E. thesis project) in PHP/Laravel.
- Designed **REST APIs** and CRUD modules for leads, sales, tasks, targets, and products.
- Optimized **MySQL** schema and queries, improving data-structure efficiency and query performance.

TECHNICAL SKILLS

AI / ML:	PyTorch, HuggingFace, CLIP, RAG, ChromaDB, LLM / Claude APIs, Mixed-Precision Training
Programming:	Python, SQL, TypeScript, JavaScript, C/C++, Java
Backend / Tools:	FastAPI, SQLAlchemy, REST APIs, Docker, Git, Playwright
Databases:	PostgreSQL, MySQL, MongoDB
Spoken Languages:	English — C1 (advanced) | German — A2 (elementary)